

VINEYARD

Wastewater Collection System Impact Fee Analysis

May 2025

ZIONS PUBLIC FINANCE, INC.

EXECUTIVE SUMMARY

An impact fee is a one-time fee imposed on new development activity to mitigate the impact of new development on capital facilities. In conjunction with this Impact Fee Analysis, Hansen Allen & Luce, Inc. prepared the Vineyard Wastewater Collection System Impact Fee Facilities Plan (IFFP) April 2025. The IFFP forms the basis for this impact fee analysis.

The recommended impact fee structure presented in this analysis has been prepared to satisfy the Impact Fees Act, Utah Code Ann. § 11-36a-101 et. seq., and represents the maximum impact fees Vineyard ("City") may assess. The City will be required to use revenue sources other than impact fees to fund any projects that constitute repair and replacement, cure any existing deficiencies, or increase the level of service for existing users.

Wastewater[1] Demand

The City is anticipated to grow by 10,598 equivalent residential units (ERUs) between 2023 and 2033 – the timeframe of the IFFP analysis.[2]

Wastewater Service Levels

Level of service (LOS) defines the wastewater capital facility demands that a typical Equivalent Residential Unit (ERU), will require as 160 gpd.

Wastewater Service Area

There is one service area for wastewater that encompasses the boundaries of Vineyard City.

Excess Capacity

According to the IFFP, there is excess capacity in the wastewater system in the amount of $3,463,047 that will serve new development over the next 10 years.

New Construction

Total new construction costs required by growth over the next 10 years are projected to reach $5,109,327.

Wastewater System Impact Fee Calculation

The gross fee is first calculated, followed by credits to be made for the portion of future projects that will benefit existing development.

TABLE 1: MAXIMUM GROSS FEE PER ERU

Summary	Amount
Existing Buy-In Costs	$326.76
Interest on Bonds	$43.10
New Construction	$482.10
Consultant Costs	$6.00

[1] The terms "sewer" and "wastewater" are used interchangeably in this analysis.
[2] IFFP, p. 2-1.

Summary	Amount
Fund Balance	($37.80)
Subtotal before Credits	**$820.17**

Credits must be made for the proportionate share of new projects that benefit existing development so that new development does not pay more than its proportionate share of costs. According to the IFFP, the cost of new projects that benefit existing development is $2,492,000. These projects are anticipated to be paid for with increased rates, spread over 10 years, and therefore new development must be credited so that it does not pay the full impact fee as well as higher rates.

Credits must also be made for future payments on the outstanding Series 2023 Water and Wastewater Revenue Bond issued in the amount of $14,350,000. Wastewater is responsible for ten percent of the bond. Credits for future payments are then subtracted from the maximum gross fee shown in Table 1 to show the maximum impact fee per year.

TABLE 2: MAXIMUM FEE PER ERU

	Gross Fee before Credits	NPV* Deficiency Projects	NPV* Outstanding Bond	Maximum Fee per ERU
2025	$820.17	$165.53	$18.39	$636.25
2026	$820.17	$140.41	$16.85	$662.91
2027	$820.17	$120.17	$15.70	$684.31
2028	$820.17	$102.35	$14.73	$703.09
2029	$820.17	$85.89	$13.89	$720.40
2030	$820.17	$70.40	$13.13	$736.65
2031	$820.17	$55.66	$12.44	$752.07
2032	$820.17	$41.36	$11.81	$767.00
2033	$820.17	$27.38	$11.23	$781.56
2034	$820.17	$13.63	$10.68	$795.86

Residential units (per door) are charged based on a ¾" meter size (1 ERC) while all nonresidential development is charged based on meter size.

TABLE 3: MAXIMUM FEE BY METER SIZE, 2025-2029

Meter Size	ERCs	2025	2026	2027	2028	2029
3/4"	1	$636.25	$662.91	$684.31	$703.09	$720.40
1"	1.67	$1,062.54	$1,107.06	$1,142.79	$1,174.16	$1,203.07
1 ½"	3.33	$2,118.72	$2,207.48	$2,278.74	$2,341.30	$2,398.93
2"	5.33	$3,391.23	$3,533.30	$3,647.36	$3,747.48	$3,839.73
3"	10	$6,362.53	$6,629.08	$6,843.07	$7,030.92	$7,204.00
4"	16.67	$10,606.33	$11,050.67	$11,407.40	$11,720.55	$12,009.07
6"	33.33	$21,206.30	$22,094.71	$22,807.95	$23,434.06	$24,010.94
8"	53.33	$33,931.35	$35,352.86	$36,494.09	$37,495.90	$38,418.94
10"	76.67	$48,781.49	$50,825.12	$52,465.81	$53,906.07	$55,233.08

TABLE 4: MAXIMUM FEE BY METER SIZE, 2030-2034

Meter Size	ERCs	2030	2031	2032	2033	2034
3/4”	1	$736.65	$752.07	$767.00	$781.56	$795.86
1”	1.67	$1,230.20	$1,255.96	$1,280.89	$1,305.21	$1,329.09
1 ½”	3.33	$2,453.04	$2,504.40	$2,554.12	$2,602.59	$2,650.22
2”	5.33	$3,926.33	$4,008.54	$4,088.13	$4,165.71	$4,241.94
3”	10	$7,366.48	$7,520.71	$7,670.03	$7,815.60	$7,958.61
4”	16.67	$12,279.92	$12,537.02	$12,785.94	$13,028.60	$13,267.01
6”	33.33	$24,552.47	$25,066.53	$25,564.21	$26,049.39	$26,526.06
8”	53.33	$39,285.43	$40,107.95	$40,904.27	$41,680.59	$42,443.28
10”	76.67	$56,478.79	$57,661.29	$58,806.12	$59,922.20	$61,018.68

Non-Standard Fee Adjustment

For development with unique characteristics and circumstances, a fee adjustment may be made by using the following formula:

Average gpd/160 gpd x the maximum fee per ERU per year

CHAPTER 1: OVERVIEW OF THE WASTEWATER IMPACT FEES

Summary

An impact fee is intended to recover the City’s costs of building wastewater system capacity to serve new residential and non-residential development rather than passing these growth-related costs on to existing users through rates. The Utah Impact Fees Act allows only certain costs to be included in an impact fee so that only the fair cost of expansionary projects or existing unused capacity paid for by the City is assessed through an impact fee.

Costs to be Included in the Impact Fee

The impact fees proposed in this analysis are calculated based upon:

- Buy-in to existing, excess capacity;
- New capital infrastructure that will serve new development; and
- Professional and planning expenses related to the construction of system improvements that will serve new development.
- Necessary credits

The costs that cannot be included in the impact fee are as follows:

- Costs for projects that cure system deficiencies;
- Costs for projects that increase the LOS above that which is currently provided;
- Operations and maintenance costs;
- Costs of facilities funded by grants or other funds that the City does not have to repay; and
- Costs of reconstruction of facilities that do not have capacity to serve new growth.

Utah Code Legal Requirements

Utah law requires that communities and special districts prepare an Impact Fee Analysis (IFA) before enacting an impact fee. Utah law also requires that communities/districts give notice of their intent to prepare and adopt an IFA. This IFA follows all legal requirements as outlined below. The City has retained Zions Public Finance, Inc. (ZPFI) to prepare this Impact Fee Analysis in accordance with legal requirements.

Notice of Intent to Prepare Impact Fee Analysis

A local political subdivision must provide written notice of its intent to prepare an IFA before preparing the Plan (Utah Code §11-36a-503). This notice must be posted on the Utah Public Notice website. The City has complied with this noticing requirement for the IFA by posting notice.

Preparation of Impact Fee Analysis

Utah Code requires that each local political subdivision, before imposing an impact fee, prepare an impact fee analysis. (Utah Code 11-36a-304).

Section 11-36a-304 of the Utah Code outlines the requirements of an impact fee analysis which is required to:

(1) An impact fee analysis shall:

- (a) identify the anticipated impact on or consumption of any existing capacity of a public facility by the anticipated development activity;
- (b) identify the anticipated impact on system improvements required by the anticipated development activity to maintain the established level of service for each public facility;
- (c) demonstrate how the anticipated impacts described in Subsections (1)(a) and (b) are reasonably related to the anticipated development activity;
- (d) estimate the proportionate share of:
  - (i) the costs for existing capacity that will be recouped; and
  - (ii) the costs of impacts on system improvements that are reasonably related to the new development activity; and
- (e) identify how the impact fee was calculated.

(2) In analyzing whether or not the proportionate share of the costs of public facilities are reasonably related to the new development activity, the local political subdivision or private entity, as the case may be, shall identify, if applicable:

- (a) the cost of each existing public facility that has excess capacity to serve the anticipated development resulting from the new development activity;
- (b) the cost of system improvements for each public facility;

(c) other than impact fees, the manner of financing for each public facility, such as user charges, special assessments, bonded indebtedness, general taxes, or federal grants;

(d) the relative extent to which development activity will contribute to financing the excess capacity of and system improvements for each existing public facility, by such means as user charges, special assessments, or payment from the proceeds of general taxes;

(e) the relative extent to which development activity will contribute to the cost of existing public facilities and system improvements in the future;

(f) the extent to which the development activity is entitled to a credit against impact fees because the development activity will dedicate system improvements or public facilities that will offset the demand for system improvements, inside or outside the proposed development;

(g) extraordinary costs, if any, in servicing the newly-developed properties; and

(h) the time-price differential inherent in fair comparisons of amounts paid at different times.

Certification of Impact Fee Analysis

Utah Code states that an Impact Fee Analysis shall include a written certification from the person or entity that prepares the Impact Fee Analysis. This certification is included at the conclusion of this analysis.

CHAPTER 2: IMPACT FROM GROWTH UPON THE CITY'S FACILITIES AND LEVEL OF SERVICE

Utah Code 11-36a-304(1)(a)

Projected Wastewater Demands

Table 5 shows ERU growth projections which will place additional demand on the City's wastewater system. The City's wastewater system serves 5,878 equivalent residential units (ERUs) which will grow to an estimated 16,476 ERUs by 2033. The IFFP is based on 2023 data and calculates the growth between 2023 and 2033 as 10,598 ERUs.

Water Service Area

ERUs within Vineyard are projected to grow as follows:

TABLE 5: GROWTH IN DEMAND

Year	ERUs
2023	5,878
2024	6,358
2025	7,462
2026	9,141
2027	10,456
2028	11,549
2029	12,598
2030	13,646
2031	14,586
2032	15,536
2033	16,476
Growth, 2023-2033	**10,598**

Source: IFFP, 3-1

Existing and Proposed LOS Analysis

Level of service (LOS) defines the wastewater capital facility demands that a typical ERU will require and should pay for with impact fees. The IFFP defines service levels as 160 gpd per ERU.[3]

Utah Code 11-36a-304(1)(b)(c)

Excess Capacity

The IFFP identifies 80 percent excess capacity in the existing system which has an actual cost of $9,516,028 for impact-fee eligible projects.[4] Of this amount, $3,463,047 represents excess capacity

[3] IFFP, p. iii
[4] Vineyard Asset List

that will be consumed by 2033, calculated as the percentage of ERU growth by 2033 (10,598 ERUs) compared to total capacity of the existing system (29,122 ERUs).

TABLE 6: EXISTING EXCESS CAPACITY

Excess Capacity	Amount
Total Excess Capacity	$9,516,028
Total Capacity ERUs	29,122
Capacity Consumed 2023-2033, ERUs	10,598
Percent of Total Capacity	36.39%
Cost of Capacity Consumed 2023-2033	$3,463,047
Source: IFFP, 3-2	

CHAPTER 4: SYSTEM IMPROVEMENTS REQUIRED FROM DEVELOPMENT ACTIVITY

Utah Code 11-36a-304(1)(b)(c)

The means by which the City will meet growth demands include constructing the following projects as set forth in the Impact Fee Facilities Plan. The cost of impact-fee eligible capital facility construction projects over the next 10 years total $5,109,327.

TABLE 7: NEW CONSTRUCTION IMPROVEMENTS

Project	Total Cost	Existing Dev	10-Yr Cost	Growth Beyond 10 Yrs
Install 4,500 ft of 6" force main	$1,147,000	$1,147,000		
Install 2,560 ft of 12" gravity line	$1,345,000	$1,345,000		
Install 1,700 ft of 10” gravity line	$858,000	$0	$391,201	$466,799
Install 2,400 ft of 15” gravity line	$1,337,000	$0	$609,599	$727,401
Install 1,200 ft of 12” gravity line and 4,600 ft of 15” gravity line	$3,193,000	$0	$1,455,834	$1,737,166
Install 4,500 ft of 8” gravity line	$2,192,000	$0	$999,433	$1,192,567
Install 1,800 ft of 8” gravity line, 1,700 ft of 10” gravity line, and 3,050 ft of 15” gravity line	$3,434,000	$0	$1,565,717	$1,868,283
Install two new pumps at Lift Station 2	$192,000	$0	$87,542	$104,458
Total	**$11,206,000**	**$2,492,000**	**$5,109,327**	**$6,096,673**

CHAPTER 5: PROPORTIONATE SHARE ANALYSIS

Maximum Legal Wastewater Impact Fee per ERU

The Impact Fees Act requires the Impact Fee Analysis to estimate the proportionate share of the future and actual cost of existing system improvements that benefit new growth that can be recouped through impact fees. The impact fee for existing assets must be based on the actual costs (when excess capacity is available) while the fees for construction of new facilities can be based on reasonable future costs of the system.

Existing Excess Capacity

The actual cost of the City's wastewater system that will serve an additional 10,598 ERUs by 2033 is $9,516,029, with $3,463,047 of the cost of the excess capacity allocated to growth between 2023 and 2033. Therefore, the buy-in cost is $326.76 per ERU.

TABLE 8: EXCESS CAPACITY PROPORTIONATE SHARE CALCULATION

Excess Capacity	Amount
Total Excess Capacity	$9,516,028
Total Capacity ERUs	29,122
Capacity Consumed 2023-2033, ERUs	10,598
Percent of Total Capacity	36.39%
Cost of Capacity Consumed 2023-2033	$3,463,047
Growth in ERUs, 2023-2033	10,598
Buy-In Cost per ERU	**$326.76**

Interest costs can be included for the Series 2023 Water and Sewer Revenue Bonds which are used to fund existing facilities. The total interest cost on the bond is $12,551,493. Wastewater is responsible for ten percent of the interest cost which is spread among the total capacity (29,122 ERUs) served by the projects funded by the bond.

TABLE 9: INTEREST COST CALCULATION

Interest Calculation on Outstanding Bond	
Total Interest	$1,255,149
Capacity ERUs of Outstanding Bond	29,122
Interest Cost per ERU	**$43.10**

New Construction

New construction projects estimated to benefit new development within the next 10 years total $5,109,327. When divided by the projected growth in ERUs, this results in a cost of $482.10 per ERU.

TABLE 10: NEW CONSTRUCTION PROJECTS

New Improvements	Amount
Cost to 10-Year Growth	$5,109,326.62
Growth in ERUs, 2023-2033	10,598
Cost per ERU	**$482.10**

Consultant Costs

Consultant costs are an allowable expense and result in a proportionate share cost of $6.00 per ERU.

TABLE 11: CONSULTANT COSTS

Consultant Cost	
Consultant Costs	$63,640
Growth in ERUs, 2023-2033	10,598
Cost per ERU	**$6.00**

Fund Balance

The fund balance of $400,588.86[5] can be used to offset some of the construction costs of new projects shown in Table 7. A credit of $37.80 per ERU is therefore applied to the new construction cost per ERU.

TABLE 12: IMPACT FEE FUND BALANCE

Impact Fee Fund Balance	Amount
Fund Balance	$400,588.86
Growth in ERUs, 2023-2033	10,598
Credit per ERU	**($37.80)**

Summary of Gross Impact Fee Before Credits

The gross impact fee includes buy-in costs for existing, excess capacity, interest cost on the outstanding bond, new construction costs (less fund balance) and consultant costs.

TABLE 13: IMPACT FEE FUND BALANCE

Summary	Amount
Existing Buy-In Costs	$326.76
Interest on Bonds	$43.10
New Construction	$482.10
Consultant Costs	$6.00
Fund Balance	($37.80)
Subtotal before Credits	**$820.17**

Credits Against Impact Fees

Credits must be made for the proportionate share of new projects that benefit existing development as well as the Series 2023 Water/Wastewater Revenue Bond so that new development is not charged more than its fair share.

[5] Source: Vineyard City

The proportionate share of new projects benefitting existing development is $2,492,000 based on the IFFP which is anticipated to be paid for with increased rates, spread over 10 years, and therefore new development must be credited so that it does not pay the full impact fee as well as higher rates.

TABLE 14: CREDITS ON PROJECTS BENEFITTING EXISTING DEVELOPMENT

Year	ERUs	Payment	Payment per ERU	NPV* - Deficiency Projects
2025	7,462	$249,200	$33.40	$165.53
2026	9,141	$249,200	$27.26	$140.41
2027	10,456	$249,200	$23.83	$120.17
2028	11,549	$249,200	$21.58	$102.35
2029	12,598	$249,200	$19.78	$85.89
2030	13,646	$249,200	$18.26	$70.40
2031	14,586	$249,200	$17.08	$55.66
2032	15,536	$249,200	$16.04	$41.36
2033	16,476	$249,200	$15.13	$27.38
2034	17,416	$249,200	$14.31	$13.63

NPV = net present value discounted at 5 percent

There is an outstanding Series 2023 Water/Wastewater Revenue Bond issued for $14,350,000. Wastewater is responsible for ten percent of the bond costs, and existing development costs represent 20.2 percent[6] of the total costs, which must be credited against the gross impact fee.

TABLE 15: CREDITS FOR SERIES 2023 WATER/WASTEWATER REVENUE BOND

Year	P+I	Wastewater Share	Existing Share	ERUs	Payment per ERU	NPV*
2025	$907,638	$90,764	$18,320	7,462	$2.46	$18.39
2026	$906,138	$90,614	$18,290	9,141	$2.00	$16.85
2027	$904,138	$90,414	$18,249	10,456	$1.75	$15.70
2028	$906,638	$90,664	$18,300	11,549	$1.58	$14.73
2029	$908,388	$90,839	$18,335	12,598	$1.46	$13.89
2030	$904,388	$90,439	$18,254	13,646	$1.34	$13.13
2031	$904,888	$90,489	$18,264	14,586	$1.25	$12.44
2032	$904,638	$90,464	$18,259	15,536	$1.18	$11.81
2033	$903,638	$90,364	$18,239	16,476	$1.11	$11.23

*NPV = net present value discounted at 5 percent

The maximum impact fee per ERU is calculated by subtracting the two credits from the gross impact fee shown in Table 11.

[6] Calculated by dividing the 5,878 existing ERUs by the capacity served by the bond projects (29,122 ERUs).

TABLE 16: MAXIMUM FEE PER ERU

	Gross Fee before Credits	NPV* Deficiency Projects	NPV* Outstanding Bond	Maximum Fee per ERU
2025	$820.17	$165.53	$18.39	$636.25
2026	$820.17	$140.41	$16.85	$662.91
2027	$820.17	$120.17	$15.70	$684.31
2028	$820.17	$102.35	$14.73	$703.09
2029	$820.17	$85.89	$13.89	$720.40
2030	$820.17	$70.40	$13.13	$736.65
2031	$820.17	$55.66	$12.44	$752.07
2032	$820.17	$41.36	$11.81	$767.00
2033	$820.17	$27.38	$11.23	$781.56
2034	$820.17	$13.63	$10.68	$795.86

*NPV = net present value discounted at 5 percent

Residential units (per door) are charged based on a ¾" meter size (1 ERC) while all nonresidential development is charged based on meter size.

TABLE 17: MAXIMUM FEE BY METER SIZE, 2025-2029

Meter Size	ERCs	2025	2026	2027	2028	2029
3/4"	1	$636.25	$662.91	$684.31	$703.09	$720.40
1"	1.67	$1,062.54	$1,107.06	$1,142.79	$1,174.16	$1,203.07
1 ½"	3.33	$2,118.72	$2,207.48	$2,278.74	$2,341.30	$2,398.93
2"	5.33	$3,391.23	$3,533.30	$3,647.36	$3,747.48	$3,839.73
3"	10	$6,362.53	$6,629.08	$6,843.07	$7,030.92	$7,204.00
4"	16.67	$10,606.33	$11,050.67	$11,407.40	$11,720.55	$12,009.07
6"	33.33	$21,206.30	$22,094.71	$22,807.95	$23,434.06	$24,010.94
8"	53.33	$33,931.35	$35,352.86	$36,494.09	$37,495.90	$38,418.94
10"	76.67	$48,781.49	$50,825.12	$52,465.81	$53,906.07	$55,233.08

TABLE 18: MAXIMUM FEE BY METER SIZE, 2030-2034

Meter Size	ERCs	2030	2031	2032	2033	2034
3/4"	1	$736.65	$752.07	$767.00	$781.56	$795.86
1"	1.67	$1,230.20	$1,255.96	$1,280.89	$1,305.21	$1,329.09
1 ½"	3.33	$2,453.04	$2,504.40	$2,554.12	$2,602.59	$2,650.22
2"	5.33	$3,926.33	$4,008.54	$4,088.13	$4,165.71	$4,241.94
3"	10	$7,366.48	$7,520.71	$7,670.03	$7,815.60	$7,958.61
4"	16.67	$12,279.92	$12,537.02	$12,785.94	$13,028.60	$13,267.01
6"	33.33	$24,552.47	$25,066.53	$25,564.21	$26,049.39	$26,526.06
8"	53.33	$39,285.43	$40,107.95	$40,904.27	$41,680.59	$42,443.28

Meter Size	ERCs	2030	2031	2032	2033	2034
10”	76.67	$56,478.79	$57,661.29	$58,806.12	$59,922.20	$61,018.68

Non-Standard Fee Adjustment

For development with unique characteristics and circumstances, a fee adjustment may be made by using the following formula:

Average gpd/160 gpd x the maximum fee per ERU per year

CERTIFICATION

Zions Public Finance, Inc. certifies that the attached impact fee analysis:

1. includes only the cost of public facilities that are:
    a. allowed under the Impact Fees Act; and
    b. actually incurred; or
    c. projected to be incurred or encumbered within six years after the day on which each impact fee is paid;

2. does not include:
    a. costs of operation and maintenance of public facilities; or
    b. cost for qualifying public facilities that will raise the level of service for the facilities, through impact fees, above the level of service that is supported by existing residents;

3. offset costs with grants or other alternate sources of payment; and

4. complies in each and every relevant respect with the Impact Fees Act.